

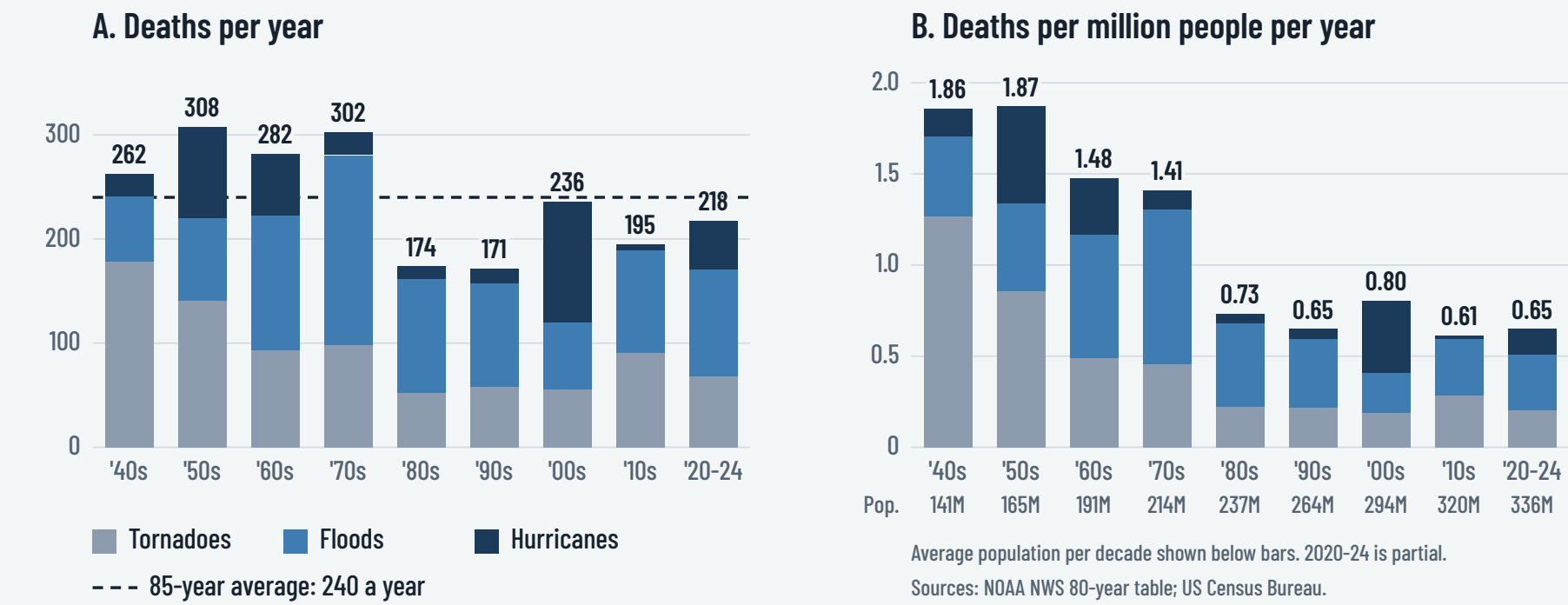
Deadlier Weather, Safer People?

Eighty-five years of US weather deaths, and why heat is harder to count

Tornadoes, floods and hurricanes kill about 240 Americans a year, close to the toll in the 1940s. The country now has more than twice as many people, so each person's risk has dropped by about two thirds. Heat is harder to judge. Its official count has nearly quadrupled since the 2010s, and the ways of counting heat deaths disagree by a factor of a thousand.

Benjamin Mahmutovic
Econ 238, University of Rochester, Fall 2026

Divide by population and the trend appears

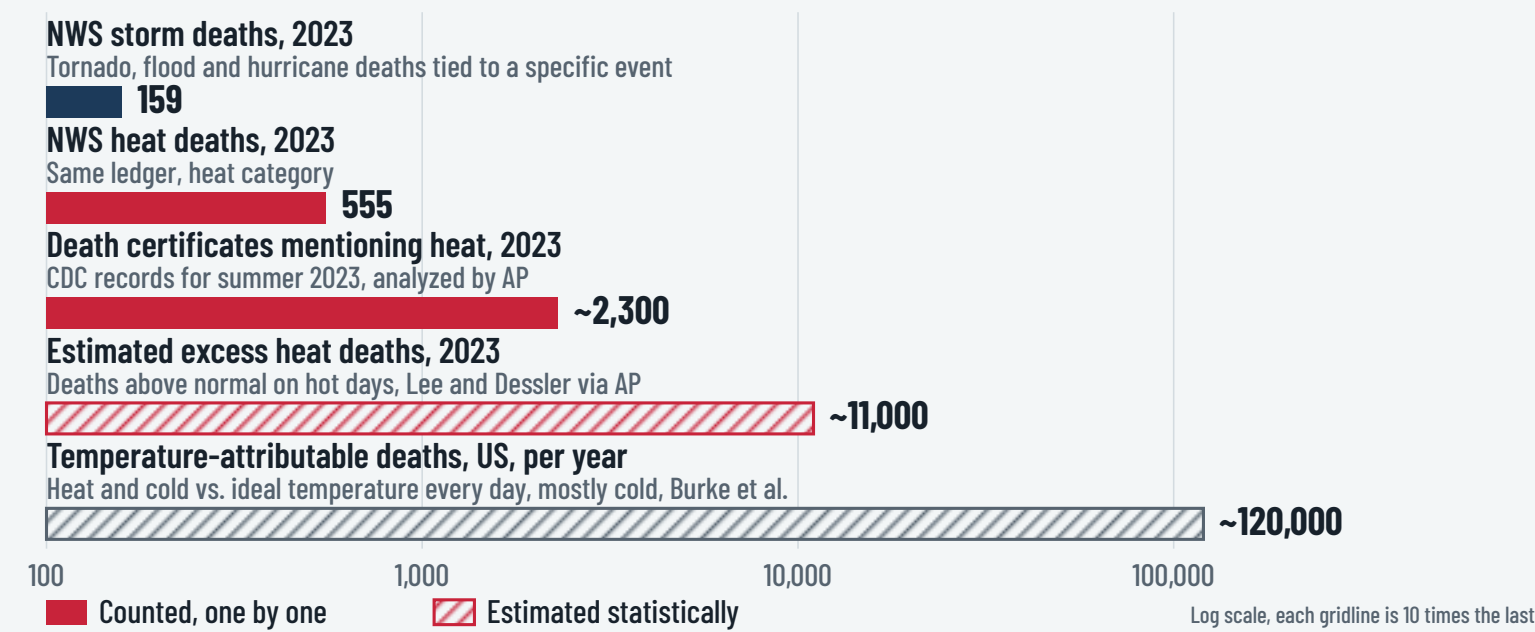


The raw counts barely move. The 1940s averaged 262 storm deaths a year, the 2010s averaged 195, and 2020 to 2024 averaged 218. Over the same period the population grew from 141 million to 336 million.

Per million Americans, the yearly death rate fell from 1.86 to 0.61.

That rate tracks an individual's risk, and it dropped by two thirds. Single disasters swing the averages. Katrina caused 1,016 of the 2000s' 1,154 hurricane deaths. Without 2005, that decade averages 134 storm deaths a year, the lowest of any.

Heat deaths depend on how you count them



The weather service counted about 115 heat deaths a year in the 2000s and 2010s, then 438 a year from 2020 to 2024. It records a death only when officials tie it to a specific event, so it misses most heat deaths.

Burke and coauthors use mortality records from 30 countries to estimate the deaths that would not happen if every day sat at the lowest-risk temperature. For the US that is about 5 percent of all deaths, near 120,000 a year. Cold causes at least five times as many of them as heat. Their number is an attribution, not a count, and it should never be compared with NWS figures.

Before you trust a weather death number, ask

Was it counted or estimated? A list of named deaths and a statistical model measure different things.

Compared with what? An ideal temperature every day is a yardstick. No policy can deliver it.

Per how many people? A rising count in a growing, aging population can still mean falling risk.

Toward heat, then a slowdown

Baylis and coauthors used Census county data for the 1950s to 2000s and IRS records for 1983 to 2018. Nationally, migrants chose counties that were already warm. Within a county, warming pushed people out. Each extra 100 cooling degree days a year cut annual net migration by 0.9 to 1.9 points.

Net domestic migration, people moving between states, Census Vintage 2025

Area	2022	2023	2025
Florida	+310,892	+183,646	+22,517
Midwest	-175,000 or worse		+16,000

July to July. Excludes births, deaths and immigration. Midwest 2022 figure covers 2021 and 2022.

Housing and insurance costs explain part of Florida's drop. I read this as a complication. We adapted well enough to move into heat, and now more people, and older ones, live where it is worst.

Exposure has a price

The IPCC concluded in 2012 that growing exposure of people and property, not a detected climate trend, drove the long-run rise in disaster losses. Exposure responds to prices. Subsidized flood insurance and federal disaster aid make it cheaper to stay in harm's way, so people keep building there.

Moving would cost little. Nordhaus put about \$600 billion of US structures low in the Atlantic hurricane zone. Rebuilding them on higher ground as they wear out, at a 15 percent premium over 50-year lifetimes, costs about \$1.8 billion a year. That is 0.006 percent of 2025 GDP.

Energy is a health input. Doubling home heating prices raises winter deaths about 4 percent. California's rules requiring water, shade and breaks for outdoor workers cut the effect of heat on job injuries by 75 to 90 percent.

But wait

The ledger changed. The NWS began counting heat in 1986 and logged 28 heat deaths a year through 1994. Nobody should believe that number. Better coroner reporting and an older population explain part of the recent rise.

Storm counts miss slow deaths. The NWS lists 7 hurricane deaths for 2017. Puerto Rico's official toll for Hurricane Maria, based on an excess-death study, is 2,975.

Storms are not the whole story. Burke and coauthors find the share of US deaths tied to temperature was the same in the 2010s as in the 1980s. I'm confident we got better at surviving storms. Whether we got better at living with temperature is still open.

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3. US Census Bureau. Vintage 2025 population estimates, release CB26-20, January 27, 2026. [census.gov/newsroom/press-releases/2026/population-growth-slows.html](https://www.census.gov/newsroom/press-releases/2026/population-growth-slows.html)

4. Baylis, Bharadwaj, Mullins and Obradovich, 2025. Climate and migration in the United States. Journal of Public Economics 249. doi.org/10.1016/j.jpubeco.2025.105446

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10. George Washington University Milken Institute SPH, 2018. Hurricane Maria excess deaths in Puerto Rico. [gwtoday.gwu.edu/gw-researchers-2975-excess-deaths-linked-hurricane-maria](https://www.gwtoday.gwu.edu/gw-researchers-2975-excess-deaths-linked-hurricane-maria)



Full story, data and methods
bimah123456.github.io/econ238-portfolio/week4/